

# CrazyFlie Workspace Documentation

## Part 1 - Workspace Setup and Hardware Connection

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# 1 Objective

The objective of this session was to build a clean ROS2 workspace for CrazyFlie development and establish successful communication with the CrazyFlie 2.1 drone.

This documentation covers

- Creating a clean ROS2 workspace
- Installing CrazySwarm2
- Building the workspace
- Understanding workspace structure
- Connecting the CrazyFlie
- Verifying communication

## 2 System Configuration

The development environment used is

- Ubuntu 22.04
- ROS2 Humble
- Python 3.10
- VS Code
- CrazyFlie 2.1
- Crazyradio PA

## 3 Creating a Clean Workspace

Instead of using an old workspace containing multiple unrelated ROS packages, a fresh workspace was created.

### 3.1 Create Workspace

```
mkdir -p ~/cf_ws/src  
cd ~/cf_ws
```

Workspace structure

```
cf_ws/  
  src/
```

## 4 Cloning CrazySwarm2

Clone the official CrazySwarm2 repository.

```
cd ~/cf_ws/src  
  
git clone https://github.com/IMRCLab/crazyswarm2.git
```

## 5 Installing Dependencies

Install all required ROS dependencies.

```
cd ~/cf_ws

rosdep install \
--from-paths src \
--ignore-src \
-r \
-y
```

## 6 Building the Workspace

Before building, source ROS2.

```
source /opt/ros/humble/setup.bash

colcon build --symlink-install
```

After compilation, source the workspace.

```
source ~/cf_ws/install/setup.bash
```

## 7 Workspace Structure

After building successfully, the workspace contains

```
cf_ws/
|
|-- build/
|-- install/
|-- log/
|
|-- src/
|   |
|   |-- crazyswarm2/
```

Meaning of each directory

- src : Source code
- build : Compilation files
- install : Executables and installed packages
- log : Build logs

## 8 Connecting the CrazyFlie

Launch the CrazyFlie server.

```
ros2 launch crazyflie launch.py
```

During launch the following components start

- CrazyFlie Server
- Motion Capture Node
- Teleoperation Node
- Joystick Node

## 9 Important Launch Messages

Typical successful messages include

- Requesting Parameters
- Param TOC Found
- Logging to /pose
- Logging to /status
- Memories Detected

These indicate that the CrazyFlie is communicating correctly.

## 10 Kalman Reset Message

The following message appears repeatedly

`ESTKALMAN: State out of bounds, resetting`

Reason

- No Flow Deck
- No Motion Capture
- No Lighthouse

Since no external localization system is connected, the onboard estimator cannot determine an accurate position.

For basic communication tests, this warning can be ignored.

## 11 Verifying Communication

Available services can be checked using

```
ros2 service list | grep cf231
```

Typical services include

- takeoff
- land
- emergency
- go<sub>to</sub>

## 12 Takeoff Service

A takeoff request can be sent using

```
ros2 service call \  
/cf231/takeoff \  
crazyflie_interfaces/srv/Takeoff \  
{group_mask:0, \  
height:0.2, \  
duration:{sec:2,nanosec:0}}
```

This sends a high-level takeoff command to the CrazyFlie.

## 13 Lessons Learned

- Always use a clean ROS workspace.
- Source the workspace before running ROS commands.
- Build using `colcon`.
- Verify services before writing ROS code.
- Keep the launch terminal running.
- Build and source after every package modification.